Small and Medium Enterprise Technology Development (R&D) Support Business R&D Plan

① Project Overview

Project Name	Development of a RAG-based Business-specific Real-time Multilingual Translation AI Solution Global Sales, Consultation, Interpretation, and Document Automation Integrated Platform for Export Revenue Innovation for SMEs
Development Goal	Implement a business-specific AI model by developing a RAG-based multilingual translation model that has learned unique product names, technical specifications, and prices for each company, and a real-time streaming interpretation (in more than 8 languages) model within seconds delay for export tasks. Develop an agent type for export document automation throughout the entire export cycle, including quotes and product pages. Build a platform that integrates unique regulatory certification formatting modules such as eCTD, KC, CE, and FCC to eliminate barriers to global entry for SMEs and realize revenue innovation.
Development Method	Construct an overwhelming entry barrier by fine-tuning domain-specific AI using an exclusive corpus of 1.1 million sentence pairs of business-specific 600 thousand regulatory certifications and 500 thousand. [AI Model Upgrading] Combine RAG technology that injects company-specific knowledge into hybrid language models based on NLLB/EXAONE to reduce hallucination phenomena by 70%. [InfiniSST and Whisper-based Technology] Secure simultaneous interpretation performance within a delay of seconds using proven AI platforms. [SaaS Platform Commercialization] Utilize the infrastructure of validated AI platforms (Oz-i) by the lead institution to develop a cloud-based SaaS web-mobile-API integrated environment that allows companies to link and use their data within 3 days without IT personnel.

Research Team Composition	[Lead] PCN Co., Ltd. (Platform/Commercialization Lead): An AI company with 25 years of experience (sales of 27 billion) owning the Oz-i AI platform and various R&D business know-how, guarantees the establishment of a SaaS platform, API integration, and cloud infrastructure commercialization. [Co-joint ①] StunningBox Co., Ltd. (AI Model/Data Lead): An AI translation data company with 15 years of specialized data set accumulation, responsible for AI translation fine-tuning and providing the core asset, exclusive corpus, and multi-agent regulatory certification formatting technology. [Co-joint ②] 3 demand companies (Field Verification): Sunlight IlluCom (automobile parts), Genuine Grip (special shoes), and Gray Whale (K-Beauty), actual export top award-winning companies, provide verification data per industry and quantitatively verify the introduction effect through A/B testing. .
Expected Effects	[Economic and Industrial Impact] Increase demand companies' export order rate by over 20%, and drastically reduce the regulatory certification document processing period from the existing 4-8 weeks to 1-2 weeks. After 5 years of commercialization, create sales of 22 billion KRW (68 billion KRW cumulative over 5 years), generate 51 new jobs, and plan to distribute to over 500 SMEs through AI vouchers and KOTRA linkage. Achieve an export substitution effect worth 44 billion KRW cumulative over 5 years. [Acquisition of World-class Technology] Achieve a translation accuracy of BLEU 65 or higher, voice interpretation delay of 2 seconds or less, and document structure preservation rate (TEDS) of 70% or more to dominate the B2B domain-specific translation market that global big tech (Google, DeepL) could not resolve.

2 Self-Diagnosis

Project Name	Global Sales, Consultation, Interpretation, and Document Automation Integrated Platform for Export Revenue Innovation for SMEs	
Confirmation Items	**Applicable**	**Not Applicable**
1. Does the current applicant (including R&D institutions, hereinafter referred to as 'applicant') not meet the application qualifications set in the business announcement etc.?		✓
2. Are there any documents among the related documents that the current applicant has not submitted as set in the business announcement etc.?		✓
3. Is the current applicant under national R&D participation restrictions?		✓
4. Does the applicant have a history of receiving sanctions such as national R&D participation restrictions due to misuse of labor costs or false transactions?		✓
5. Is the current applicant a bankrupt, suspended, or closed company?		✓
6. Has the current applicant been subjected to delinquency dispositions on national taxes, local taxes, customs duties by tax authorities, or has information been registered as delinquency of fines, industrial accident/employment insurance by the Comprehensive Credit Information Concentration Agency?		✓
7. Is the current applicant listed on the debtor's list under the Civil Execution Act or registered as a debtor to the Comprehensive Credit Information Concentration Agency?		✓
8. Has the current applicant applied for the commencement of bankruptcy, rehabilitation procedures, personal rehabilitation procedures in court?		✓
9. Is the recent applicant in a state of debt ratio of 1,000% or more based on the recent year's settlement, or in a state of total capital erosion?		✓
Has the result of self-inspection on whether the applied project falls under the security task classification criteria of Article 45 (1) of the Enforcement Decree of the Innovation Act determined it as a security task?		✓

※ In addition to the self-diagnosis items specified above, based on the results of eligibility requirements specified in the business announcement and related regulations, it may be excluded or canceled from selection ※ The 'current' criteria for confirmation items mentioned above is based on the closing date of submissions or the point set in the business announcement and related regulations to confirm eligibility requirements ※ Each lead and joint institution should write separately

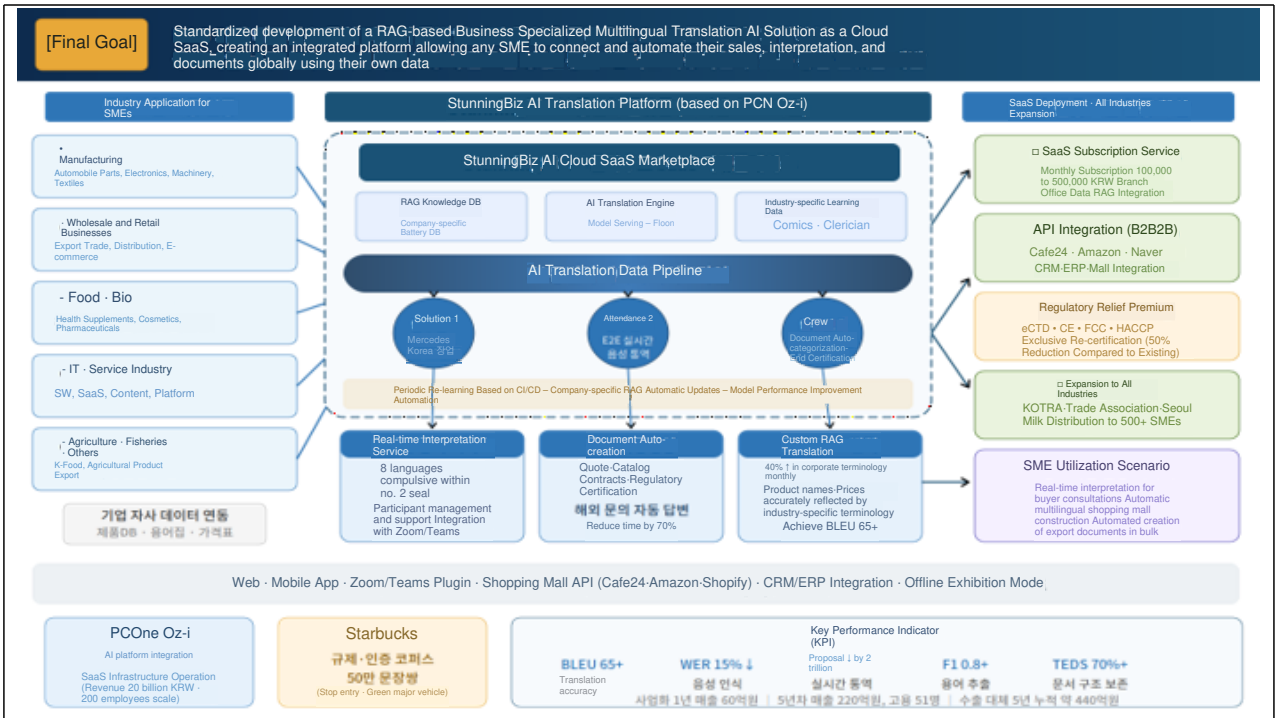
The applicant confirms the facts above and has no objection to being excluded from support or having the selection canceled if there is any deviation from the truth.

2026 July 01 day

Lead Institution Applicant : 엄 상 현 Western 문

Lead Institution Corporate Name : PCN Co., Ltd.

1. Technology Development Goals



1) Background and necessity of business promotion

(1) Environmental analysis

Export SMEs face complex barriers in overseas business due to translations that do not reflect the company-specific context, interpreting structures unfit for real-time consultations, and high-cost, long-term regulatory certification document responses. Therefore, beyond simple translation functions, an AI platform specialized in domain connecting the entire export task lifecycle based on company data and technical terminology is needed.

Core barriers	Field issues	Need for technology development
① Business context·Technical terminology	Generic AI does not sufficiently reflect company-specific product names·technical specifications·prices·contract terms, creating the risk of critical information mistranslation in negotiation and contract processes.	Company-specific product DB·price list·glossary connected with RAG for real-time reflection in translation·interpretation·document creation
② Real-time interpretation performance	The existing 3-step structure of STT→MT→TTS has a cumulative delay of more than 2-5 seconds, with limitations on background noise·speaker separation·technical terminology handling	Implementation of sub-2-second interpretation through E2E streaming interpretation architecture with speaker separation, technical terminology injection
③ Regulatory certification documents	CE·eCTD·FCC·HACCP, etc., have complex country-specific formats and document structures, with preparation periods and outsourcing costs that are a significant burden on SMEs.	Support for submission document creation by automating the translation·structuration·formatting of regulatory documents, under expert review (HITL)

(2) Market growth and urgency of development

◦ The global business AI translation market is projected to grow from approximately 1.8 billion USD in 2025 to over 7 billion USD in 2030, with rapid expansion of demand for business domain specialization.